MZUZU UNIVERSITY

FACULTY OF INFORMATION SCIENCE AND COMMUNICATIONS

DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY

ICT 4703: NETWORK ADMINSTRATION AND INFORMATION SECURITY

END OF SEMESTER EXAMINATION

DATE: JANUARY 2016 **TIME ALLOWED**: 3 HOURS

INSTRUCTIONS:

i. Answer **ALL** questions
ii. This paper contains **FIVE** pages
iii. Write your answers clearly
iv. Marks for each question are indicated
v. There are **THREE** Sections: A, B and C. Please Check

SECTION A : MULTIPLE CHOICE QUESTIONS [10 Marks]

1. Which of the following correctly describes the characteristics of switched and routed data flows?

   A. Switches create a single collision domain and a single broadcast domain,Routers provide seperate collision domains.
   B. Switches create separate collision domains and seperate broadcast domains, Routers provide seperate collision domains and separate broadcast domain as well.
   C. Switches create separate collision domains and seperate broadcast domains, Routers provide seperate collision domains.
   D. Switches create a single collision domain and seperate broadcast domains,Routers provide seperate broadcast domain as well.
   E. Switches create seperate collision domains and a single broadcast domain, Routers provide seperate broadcast domain.

2. Which of the following is a true statement about the function of a bridge ?

   A. A bridge creates or maintains a table of the ava ilable routes and their conditions.
   B. A bridge passes packets outside of its network segment if the IP address cannot be found on its table.
   C. A bridge sends the frame towards the destination host after it looks up the frame's destination in its address table.
   D. A bridge creates or maintains a table of the IP addresses of the hosts connected to its network segment.

3. Which of the following symbols in the router prompt indicates that you are in privileged mode?

   A. %
   B. #
   C. Privileged
   D. >
   E. ^

4. You are configuring a cisco switch. As you try to enter a command, you keep getting the % incomplete error response. How can you get help on the command that you are trying to enter?

   A. Reenter the command followed by a question mark to view keywords.
   B. Type " Help" followed by the command to see command parameters.
   C. Use the up arrow to review the prompt before the error.
   D. Type "?" to display all the console commands.
   E. Type history to review the prompt before the error.

5. How do you configure the line console password to "cisco" ?
   A. Line console 0, Login, Password Cisco.
   B. Line console 0 4, Login, Password cisco.
   C. Line console, Login, Password cisco.
   D. Line console 0, Login, Password cisco.

6. You are required to backup your IOS to a network location. Which of the following command will accomplish this?
   A. copy tftp flash.
   B. copy flash ftp.
   C. copy ftp flash.
   D. copy flash tftp.

7. Which layer provides segmentation of data and reassembling at the other end ?
   A. Session.
   B. Transport.
   C. Datalink.
   D. Network.

8. What command will enable an encrypted password which cannot be seen in clear text in the running config ?
   A. Enable password.
   B. Enable encrypted password.
   C. Service password encrypted.
   D. Enable secret password.

9. What command will enable a network on an interface ?
   A. Ip network 10.1.2.0.
   B. Network 10.1.0.0.
   C. Network Interface 10.1.0.0 255.255.0.0.
   D. Network 10.0.0.0.

10. You are given an ip adress 132.15.136.2/18 what subnet is the host ip on ?
   A. 132.15.136.0.
   B. 132.15.128.0.
   C. 132.15.192.0.
   D. 132.15.17.0.

SECTION B: STRUCTURED QUESTIONS

1. Remote access is often used by system administrators in large organisations to remotely control the computers on the organisation's local area network. Explain how malicious remote access administration tools (RATs) can be used. **[5 Marks]**

2. Encryption is not security. Argue. **[5 Marks]**

3. Describe **TWO** major security risks associated with wireless local area networks and measures that should be taken to reduce or eliminate them. **[10 Marks]**

4. Explain how email can be used to the recipients' disadvantage. **[5 Marks]**

5. You work in a multi-branched company. You have been given a task to assist management in trimming unnecessary ICT-related costs (Travel expenses, phone) as much as possible. As a Network Administrator with sound and updated knowledge, what would you do? **[10 Marks]**

6. Explain how Cryptography achieves the security goals. **[10 Marks]**

7. Suppose Alice wants to send "love u" to Bob using shift cipher using key=15. Calculate the cipher text that Bob will receive. **[5 Marks]**

SECTION C : PRACTICAL (50 Marks)

TechNix Malawi Ltd. Network requirements

You have been given the network address 10.51.0.0/24 to configure and provide the IP addressing for the network shown in Figure 2. The network has the following addressing requirements:

- The LILONGWE LAN 1 will require 20 host IP addresses.
- The LILONGWE LAN 2 will require 15 host IP addresses.
- The MZUZU LAN 1 will require 30 host IP addresses.
- The MZUZU LAN 2 will require 10 host IP addresses.
- The HQ-BT LAN will require 50 host IP addresses.
- Mozambique LAN will require 49 Addresses.
- Tanzania LAN will require 27 Addresses.
- The links between routers will require an IP Address at each end.

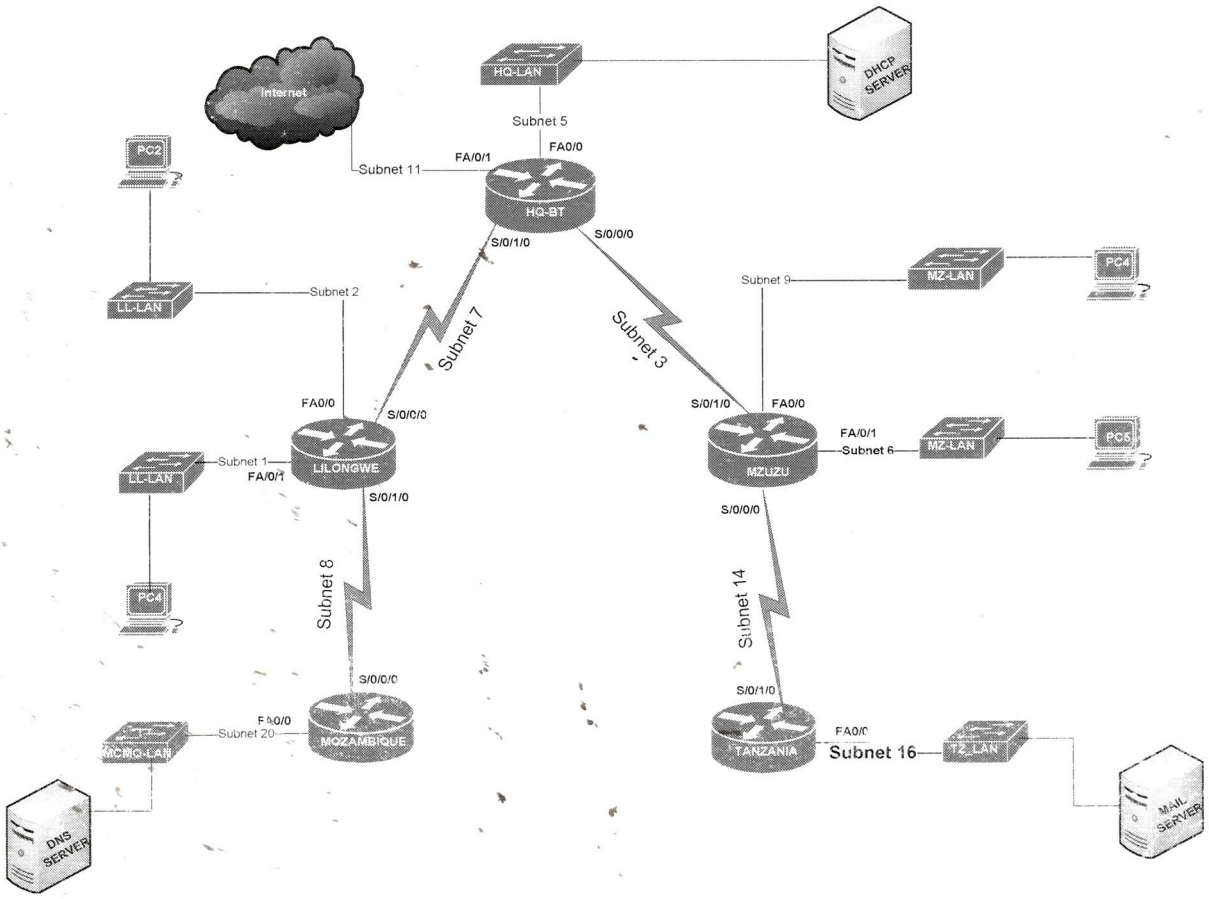


Figure 2. TechNix Malawi Ltd. Network Infrastructure

a) Perform basic configuration on the routers and switches including MOTD. **[10 Marks]**

b) Assign the appropriate addresses to the device interfaces using the following instructions: **[15 Marks]**

   i. Assign the first valid address to the serial interfaces s0/0/0 and the second valid address to S0/1/0 on each subnet between routers.
   ii. Assign the last valid address to all PCs.
   iii. Assign the third valid address to all the servers.

c) Implement Static routing on the network. **[10 Marks]**

d) Cofigure the default route on each router **[5 Marks]**

e) What command would enable OSPF? **[5 Marks]**

f) How would you backup config on a remote tftp server with an address 10.168.0.100? **[5 Marks]**

END OF QUESTION PAPER